

State of the Art in Rare-Earth-Free Permanent Magnet Technologies: 2026 Exhaustive Research Report

Executive Summary

The global transition toward sustainable energy paradigms, characterized by the mass electrification of transportation infrastructure, the proliferation of utility-scale wind-based power generation, and the ubiquitous integration of high-density robotics, has fundamentally catalyzed the demand for high-performance permanent magnets. Historically, the optimal performance metrics for these critical applications—specifically, maximal energy product $(BH)_{max}$, superior intrinsic coercivity (H_c), and substantial remanent magnetization (M_r)—have been monopolized by rare-earth-based intermetallic compounds. The most prominent

among these are Neodymium-Iron-Boron ($\text{Nd}_2\text{Fe}_{14}\text{B}$) and Samarium-Cobalt ($\text{Sm}_2\text{Co}_{17}$) systems.¹ However,

the strategic vulnerability of rare-earth element (REE) supply chains has emerged as a severe geopolitical and macroeconomic bottleneck in the mid-2020s.

As of the current operational landscape in 2026, the extraction, processing, and refining of rare-earth elements remain heavily concentrated. A single geopolitical entity controls nearly 80% of the world's rare earth extraction and 87% of global processing capacity, effectively commanding a monopoly over the highest-value nodes of the global clean-energy supply chain.³ In response to this critical vulnerability, state actors have initiated aggressive supply chain reshoring efforts, such as the U.S. Department of Defense's invocation of the Defense Production Act to subsidize domestic refining floors.⁵ Yet, expanding traditional rare-earth mining operations presents profound environmental hurdles. The extraction of elements such as Neodymium (Nd), Dysprosium (Dy), and Terbium (Tb) entails severe ecological degradation, requiring the displacement of massive volumes of earth to yield negligible quantities of active magnetic material, while simultaneously generating highly toxic, acidic, and radioactive byproducts.⁴

Consequently, the materials science, quantum chemistry, and applied physics communities have aggressively pursued the development of "Critical-Material-Free" (CMF) or rare-earth-free permanent magnets. This exhaustive report chronicles the defining scientific breakthroughs realized up to the year 2026 in the synthesis, physical characterization, and commercial scalability of these alternative magnetic systems. The core challenge in developing rare-earth-free magnets lies in compensating for the loss of the highly localized, highly anisotropic 4f electron orbitals inherent to rare earths, which generate massive spin-orbit coupling. Without rare earths, metallurgists must engineer 3d transition metals (such as iron, cobalt, and manganese) to exhibit high magnetocrystalline anisotropy energy (MAE). This is

achieved through aggressive structural symmetry breaking, the introduction of interstitial atoms, dynamic nanostructuring, and the harnessing of heavy, high-Z non-rare-earth elements to artificially induce spin-orbit coupling.

This analysis details nuanced developments across several highly promising material classes.

These include the forced atomic ordering of the cosmic $L1_0$ -FeNi phase (tetrataenite); the stabilization of giant saturation magnetization in submicron interstitial iron nitride (α'' -Fe₁₆N₂); the high-temperature resilience of engineered Manganese-Bismuth (MnBi) and its newly discovered Co-Bi/MnBi₂ cousins; the microstructural defect optimization of Manganese-Aluminum (MnAl); and the radical disruption of cubic symmetry in novel high-entropy borides. Through rigorous examination of validated scientific publications, this report synthesizes the mechanistic pathways, thermodynamic barriers, and industrial implications of a post-rare-earth magnetic economy, providing an exhaustive catalog of discoveries, publication links, mechanisms, and contact architectures for principal investigators.

1. The $L1_0$ -FeNi (Tetrataenite) Paradigm: Overcoming Cosmic Kinetic Barriers

Tetrataenite is an intermetallic phase of iron and nickel (FeNi) characterized by a highly ordered face-centered tetragonal ($L1_0$) crystal structure belonging to the space group $P4/mmm$. In this mathematically precise ordered state, alternating monatomic layers of iron and nickel atoms stack sequentially along the crystallographic c-axis.¹ This structural asymmetry disrupts the high symmetry of the chemically disordered, cubic close-packed (fcc) phase of FeNi, which is known meteoritically as taenite. In the disordered cubic state, the material exhibits isotropic magnetic properties and functions as a soft magnet. However, when forced into the tetragonal $L1_0$ configuration, the reduced symmetry yields a substantial uniaxial magnetocrystalline anisotropy energy. Theoretical calculations demonstrate that pure $L1_0$ -FeNi is capable of supporting a maximum magnetic energy product, $(BH)_{max}$, exceeding 42 MGOe (335 kJ/m³), positioning it as a direct competitor to mid-grade NdFeB magnets while relying exclusively on some of the Earth's most abundant and inexpensive transition metals.⁷ The primary barrier to exploiting tetrataenite industrially is its extreme kinetic formation threshold. The chemical ordering from the high-symmetry fcc phase to the lower-symmetry $L1_0$ phase is thermodynamically favored only below a critical order-disorder transition temperature, widely accepted to be approximately 320 °C (593 K).⁸ At these depressed temperatures, the solid-state atomic diffusion of iron and nickel is extraordinarily sluggish. Calculations indicate atomic mobilities on the order of roughly one atomic jump per 2,600 years.⁸ Consequently, the formation of highly ordered bulk tetrataenite in nature is exclusively

restricted to the ultra-slow cooling cores of iron-nickel meteorites, which experience cooling rates of approximately 1 Kelvin per million years over billions of years in the vacuum of space.¹

1.1 The Phosphorus Controversy and First-Principles Clarification

Efforts to synthesize tetrataenite artificially have historically relied on extreme, non-scalable laboratory methods. The first successful synthesis, achieved by Louis Néel in the 1960s, required intense neutron irradiation of a chemically disordered FeNi sample under an applied magnetic field to artificially create lattice vacancies that accelerated atomic diffusion.⁷ For decades, achieving this ordered state via conventional bulk casting was deemed physically impossible.

In late 2022, a highly publicized breakthrough claimed that the direct bulk casting of tetrataenite was achievable on industrially practicable timescales (seconds rather than millennia) via the precise metallurgical addition of phosphorus (P).⁶ The hypothesis suggested that phosphorus acted as a potent catalyst, drastically lowering the activation energy required for the fcc to $L1_0$ phase transition.⁹ However, the scientific method's rigorous self-correction mechanisms were fully deployed by 2024. The original paper was formally retracted after an exhaustive re-evaluation of the experimental data revealed fundamental misinterpretations of the crystallographic and magnetic signatures.¹¹ High-resolution X-ray diffraction re-analysis

demonstrated that the Bragg peaks initially attributed to the $L1_0$ superlattice were, in fact,

artifacts generated by the presence of iron-nickel phosphides, specifically $(\text{Fe,Ni})_3\text{P}$.⁷

Furthermore, extra Bragg reflections observed in transmission electron diffraction—previously considered definitive proof of tetrataenite superlattice reflections—were conclusively mapped to the surface oxidation of the sample during preparation.¹¹ Most damningly, highly sensitive measurements of magnetic hysteresis failed to detect the localized regions of high coercivity that must necessarily exist if the highly anisotropic tetrataenite phase were present.¹¹

Following this retraction, the theoretical community stepped in to resolve conflicting reports regarding the maximum theoretical anisotropy of the material. A comprehensive 2023 study published in the *Journal of Applied Physics* by Christopher D. Woodgate, Laura H. Lewis, and Julie B. Staunton utilized a first-principles electronic structure framework to model the

magnetocrystalline anisotropy energy of atomically ordered $L1_0$ FeNi.¹² Analyzing both

equiatomic $\text{Fe}_{0.5}\text{Ni}_{0.5}$ and an iron-rich composition ($\text{Fe}_{0.56}\text{Ni}_{0.44}$), the

computational models confirmed that the leading-order anisotropy coefficient, K_1 , heavily

dominates the higher-order coefficients K_2 and K_3 in pristine single crystals.¹³ The study

resolved a historical discrepancy: while theoretical calculations for pristine crystals often appeared to undershoot the high anisotropy values measured in highly complex, multiphase natural meteorites, the variance originates from the nanoscale polycrystalline, multivariant nature of experimental samples. In these physical samples, extreme microstructural constraints

generate much larger values of K_2 and K_3 than expected *a priori*.¹³ This validated that if the $L1_0$ phase can be manufactured, its theoretical coercivity ceiling is sufficiently high for commercial permanent magnet applications.

1.2 Mechanochemical Synthesis: Stress-Magnetic Field Annealing

With chemical doping (phosphorus) invalidated as a shortcut, researchers turned to manipulating the fundamental thermodynamic drivers of diffusion. A verified breakthrough in bulk synthesis was detailed in 2024 by a team led by Laura H. Lewis at Northeastern University, in collaboration with Plamen S. Stamenov at Trinity College Dublin, published in *Advanced Science*.³

Bypassing traditional thermal-only limitations, the research demonstrated that the application of simultaneous mechanical stress and a modest magnetic field during an extended annealing process could artificially induce significant atomic mobility.⁷ The experimental protocol subjected bulk precursor samples—which consisted entirely of the chemically disordered cubic FeNi alloy—to these dual fields for a continuous period of six weeks.¹⁴ To accurately quantify the resulting phases without destroying the sample, the team utilized simultaneous conversion

X-ray and backscattered γ -ray ^{57}Fe Mössbauer spectroscopy.¹⁴

The results were groundbreaking: the researchers confirmed the production of up to 22 volume percent of the tetragonal tetrataenite phase within the bulk matrix.¹⁴ The data extracted from the processed alloys were meticulously validated against Mössbauer parameters derived from actual natural meteoritic tetrataenite. While the meteoritic samples still exhibit a substantially higher degree of total atomic order (consistent with a uniaxial magnetocrystalline anisotropy energy of approximately 1 kJ/m³), this validated mechanochemical synthesis marks the first highly controlled, reproducible trajectory toward industrial fabrication without relying on irradiation.⁸ The application of an external stress field actively alters the lattice vacancy migration barriers, physically lowering the activation energy required for atoms to jump between lattice sites. Simultaneously, the applied magnetic field aligns the internal spin moments, altering the free energy landscape to thermodynamically favor the highly

anisotropic, low-symmetry $L1_0$ phase.¹⁴

1.3 The NITE Method for FeNi

Concurrent to the mechanochemical approach, another novel processing pathway emerged known as the Nitrogen Insertion and Topotactic Extraction (NITE) method.¹ This approach leverages the aggressive expansion of the crystal lattice via the temporary insertion of interstitial atoms. In the NITE process, nitrogen is forcibly inserted into the FeNi matrix, significantly expanding the unit cell and lowering the energy barriers for atomic rearrangement. While in this expanded state, the iron and nickel atoms possess sufficient mobility to reorder into the alternating $L1_0$ configuration. Subsequently, the nitrogen is topotactically extracted

from the lattice, causing the structure to collapse back down into the dense, highly ordered tetragonal phase.¹ Recent iterations of $L1_0$ -FeNi powder prepared by the NITE method have demonstrated a high saturation magnetization of 139 Am²/kg coupled with a measurable coercivity of 0.14 MA/m, proving the viability of interstitial manipulation for phase ordering.¹

Innovation: Tetrataenite ($L1_0$-FeNi) Synthesis & Modeling
Material/Technology: Bulk $L1_0$ -FeNi via Stress-Magnetic Field Annealing & First-Principles Modeling
Scientific Summary: Confirmed bulk ordering (up to 22 vol% $L1_0$ phase) utilizing simultaneous mechanical stress and magnetic fields over 6 weeks, validated via ⁵⁷ Fe Mössbauer spectroscopy. Resolved theoretical anisotropy ceilings through advanced DFT modeling of K_1 , K_2 , and K_3 coefficients. Overturns the retracted phosphorus-doping hypothesis.
Key Metrics: Yielded 22 vol% $L1_0$ phase; computational validation of strong uniaxial MAE.
Principal Investigators: Prof. Laura H. Lewis (Northeastern University), Dr. Christopher D. Woodgate (University of Bristol)
Publications / DOIs: 10.1002/adv.202302696 , 10.1063/5.0169752 ¹³
Contact Information: Prof. Laura H. Lewis

2. Extraterrestrial Validation: Shock-Metallurgy and Chang'E-6

The fundamental understanding of tetrataenite formation was profoundly disrupted and enriched in early 2026 following the rigorous analysis of lunar regolith retrieved by the Chinese National Space Administration's Chang'E-6 (CE-6) mission.²¹ The CE-6 mission achieved a historic milestone in 2024 by returning 1,935.3 grams of highly pristine soil from the Apollo basin, a crater nested deep within the Moon's vast South Pole-Aitken (SPA) basin.²¹ This represents the first-ever physical sample return from the lunar farside, a region defined by extreme antiquity and complex impact-related geology.²¹

A massive, multi-institutional research consortium led by Professor Yang Li and Dr. Rui Li at the

Institute of Geochemistry, Chinese Academy of Sciences, reported the groundbreaking discovery of pristine tetrataenite within these samples. Their findings, published in the premier international journal *Planet* in early 2026, represent a watershed moment for both planetary science and metallurgical engineering.²¹

2.1 Resolving the Lunar Magnetic Anomaly

For decades, orbital surveys have consistently detected patchwork networks of strong magnetic anomalies scattered across the Moon's surface, particularly on the farside within the SPA basin.²¹ This presented a severe paradox: unlike Earth, the Moon lacks a molten iron core capable of generating a global magnetic dynamo. Therefore, the presence of strongly magnetized crustal regions implied the existence of highly stable, high-coercivity localized magnetic mineral carriers.²¹ Until the CE-6 sample return, direct physical evidence of these "magnetic memories" was entirely lacking.²¹

Utilizing an array of advanced metrological techniques, including focused ion beam (FIB) sample preparation and high-resolution transmission electron microscopy (HRTEM), the research team identified discrete grains of tetrataenite deeply embedded within iron-sulfide

(troilite) matrices.²¹ The presence of this $L1_0$ phase is significant because tetrataenite is a hard magnetic mineral with exceptional remanence-carrying capacity.²¹ Previous studies on meteorites demonstrated that millimeter-scale samples containing approximately 40% tetrataenite can magnify localized coercivity and saturation remanent magnetization by more than a factor of ten.²⁵ The identification of this highly anisotropic mineral on the lunar surface directly explains how localized magnetic patterns can persist over billions of years despite the harsh radiation environment of the lunar surface.²²

2.2 The Space Weathering Monotectoid Mechanism

Beyond resolving planetary mysteries, the CE-6 discovery provided materials scientists with a

verified mechanism for rapid $L1_0$ synthesis. The researchers deduced a highly complex "space weathering" mechanism responsible for the formation of the lunar tetrataenite.

According to the analysis, the original precursor material was nickel-rich chondritic metal delivered to the lunar surface during ancient, cataclysmic impact events.²⁵

Subsequent, repetitive micrometeorite impacts over millions of years subjected this material to extreme kinetic and thermal shock. The impacts flash-heated troilite droplets coated with chemically disordered taenite (the fcc phase of FeNi).²⁵ These shockwaves spalled the droplets from their parent bodies, launching them across the lunar vacuum before they deposited back into the Apollo basin.²⁵ During the ensuing, extremely rapid cooling phase in the vacuum of space, the intense residual thermal energy and extreme pressure transients induced a monotectoid reaction.²⁵ A monotectoid reaction is a solid-state phase transformation where a single solid phase decomposes into two distinct solid phases. In this highly energetic environment, the high-temperature face-centered cubic taenite decomposed, forcing the atoms to order into body-centered tetragonal tetrataenite while simultaneously precipitating

excess iron as sub-micron-sized metallic iron particles.²⁵ For industrial materials science, the CE-6 discovery is revolutionary. It empirically validates that localized, high-energy, high-pressure impact shocks—analogueous to terrestrial processes like severe mechanical alloying, explosive compaction, or rapid thermal processing—can provide sufficient activation energy to overcome the kinetic barriers of $L1_0$ ordering without requiring million-year cooling epochs.²⁵ It provides a "cosmomimetic" blueprint for engineers to design severe plastic deformation or shock-compaction manufacturing pipelines for rare-earth-free super-magnets.

Innovation: Lunar Tetrataenite & Shock Metallurgy
Material/Technology: Discovery of $L1_0$ -FeNi in Chang'E-6 Lunar Samples
Scientific Summary: First definitive identification of tetrataenite on the lunar farside. Analysis revealed a space weathering mechanism where impact shocks triggered a monotectoid reaction in taenite, driving $L1_0$ atomic ordering and sub-micron iron precipitation during rapid cooling. Provides terrestrial proof-of-concept for shock-induced ordering.
Key Metrics: Resolves farside magnetic anomalies; establishes ultra-fast impact-shock ordering pathway for highly anisotropic FeNi.
Principal Investigators: Prof. Yang Li, Dr. Rui Li (Institute of Geochemistry, Chinese Academy of Sciences)
Publications / DOIs: 10.15302/planet.2026.26009 ²⁴
Contact Information: Prof. Yang Li

3. Interstitial Iron Nitrides (α'' -Fe₁₆N₂): Pushing the Boundaries of Magnetic Flux

While tetrataenite offers a structurally driven pathway to high coercivity, interstitial iron nitrides—specifically the elusive α'' -Fe₁₆N₂ phase—have captured intense industrial attention due to their unprecedented saturation magnetization. Originally observed in the tempering of iron-nitrogen martensite by K.H. Jack in 1950, and later synthesized in thin films by M. Takahashi in 1994, the material was surrounded by decades of fierce debate regarding the fundamental physics of its magnetic moment.³¹ However, advanced metrology in the 2020s

definitively resolved these controversies. The α'' -Fe₁₆N₂ phase is now universally recognized by the scientific community for exhibiting a "giant saturation magnetic flux density" of up to 2.9 Tesla, an upper bound that significantly surpasses the magnetic flux density of even the strongest commercial NdFeB magnets.¹

The underlying physics of this giant magnetization are best described by a sophisticated cluster+atom model.³³ Advanced characterization using polarized neutron reflectivity, polarized neutron diffraction, and X-ray magnetic circular dichroism (XMCD) confirms that the enhanced magnetic moment is a direct consequence of interstitial lattice expansion.³³ When nitrogen atoms are precisely intercalated into the iron lattice, they occupy specific interstitial sites. This physical insertion strains and expands the distance between iron atoms. The expanded lattice alters the electronic band structure, specifically increasing the degree of localization of the iron 3d electron spins, thereby massively amplifying the net magnetic moment of the cluster.³³

Beyond sheer magnetic strength, the α'' -Fe₁₆N₂ compound demonstrates a highly favorable magnetic anisotropy constant (1.8 MJ/m³) and possesses a profound thermal advantage over rare-earth magnets. Its coercivity temperature coefficient is incredibly stable, measuring approximately -0.4 Oe/°C within the operational range of 27–152 °C.³¹ This degradation rate is two full orders of magnitude lower than that of commercial NdFeB magnets

(which exhibit severe coercivity losses, roughly -81.9 Oe/K), making iron nitrides extraordinarily attractive for low-to-medium temperature continuous-duty applications like electric motors, wind turbine generators, and high-fidelity audio equipment.³¹

3.1 Submicron Powder Synthesis via CaH₂ Desiccation

Despite its exceptional intrinsic properties, synthesizing phase-pure bulk α'' -Fe₁₆N₂ has historically resulted in devastatingly low yields. The primary obstacle is diffusion kinetics: uniformly inserting nitrogen deep into the iron lattice without over-nitriding the surface and triggering phase separation into non-magnetic, stable nitrides (such as Fe₄N) is incredibly difficult.

In January 2025, a critical synthesis milestone was published in the Royal Society of Chemistry's *Materials Advances* journal by a team led by Takayuki Tsuchida and Hirotsugu Takizawa at Tohoku University.³⁴ The researchers deduced that the efficiency and uniformity of the ultimate nitridation process are fundamentally dependent on the exact crystallite size of the precursor α -Fe nanoparticles. If the intermediate iron grains are too large, nitrogen gas cannot diffuse completely to the crystalline core during the nitridation step, yielding heterogeneous, multiphase particles with diluted magnetic properties.³⁴

In the traditional industrial synthesis route, the precursor material, magnetite (Fe₃O₄), is reduced to pure α -Fe using hydrogen gas (H₂) at elevated temperatures (typically 400 °C or higher). However, the chemical byproduct of this reduction reaction is water vapor

(H₂O). The thermodynamic consequence of a high partial pressure of water vapor within the reaction chamber is that it acts as a powerful catalyst for rapid nucleation and grain coarsening, causing the iron nanoparticles to rapidly fuse and expand beyond the optimal diffusion threshold.³⁴

To engineer a thermodynamically suppressed growth environment, Tsuchida's team introduced a highly reactive calcium hydride (CaH₂) drying agent directly into the H₂ reduction system.³⁴ The CaH₂ aggressively and continuously absorbed the generated water vapor, drastically lowering its partial pressure inside the reactor. This profound chemical intervention

altered the activation energy of the reduction phase, allowing the reduction of Fe₃O₄ to proceed at a significantly depressed temperature of 300 °C.³⁴

The combination of lower thermal energy and an ultra-arid environment effectively halted grain coarsening. This precise environmental control yielded highly uniform α' -Fe precursor powders with crystallite diameters restricted to precisely 20 nanometers.³⁵ Upon subsequent low-temperature nitridation at 140 °C, these geometrically optimized precursor nanoparticles allowed for complete, uniform nitrogen diffusion across the entire particle volume.³⁵ The result was the synthesis of a submicron-sized powder (composed of primary particles ranging from 20–30 nm in diameter) boasting an extraordinary 97 wt% yield of the pure, highly magnetic α'' -Fe₁₆N₂ phase.³⁵

These phase-pure particles exhibited significantly enhanced coercivity compared to untreated α' -Fe equivalents, confirming the successful generation of high magnetocrystalline anisotropy alongside the giant saturation magnetization.³⁷ This highly scalable, low-temperature chemical route provides the necessary pristine feedstock required for macroscopic commercial consolidation efforts. It directly supports the ambitious industrial upscaling programs of commercial entities like Niron Magnetics, which are currently pioneering the mass deployment of these "Clean Earth Magnets" as direct drop-in replacements for vulnerable rare-earth supply chains.³¹

Innovation: Interstitial Iron Nitride (α''-Fe₁₆N₂)
Material/Technology: High-Yield Submicron α'' -Fe ₁₆ N ₂ via CaH ₂ Desiccation
Scientific Summary: Achieved 97 wt% phase purity of the giant saturation magnetization phase (α'' -Fe ₁₆ N ₂). By incorporating a CaH ₂ drying agent during the H ₂ reduction of Fe ₃ O ₄ , water vapor partial pressure was suppressed, allowing the reduction temperature to drop to 300 °C. This restricted

intermediate α -Fe grain growth to an optimal 20 nm, enabling total and uniform nitrogen diffusion during a subsequent 140 °C nitridation step.
Key Metrics: 97 wt% pure α'' -Fe ₁₆ N ₂ phase; highly stable coercivity temperature coefficient ($\approx$ 0 Oe/°C).
Principal Investigators: Dr. Takayuki Tsuchida, Prof. Hirotugu Takizawa (Tohoku University)
Publications / DOIs: (https://doi.org/10.1039/D4MA00961D) ³⁴
Contact Information: Dr. Takayuki Tsuchida

4. The Bismuth Revolution: Engineering High-Temperature Resilience in MnBi

For advanced engineering applications demanding extreme thermal resilience—specifically high-torque traction motors in heavy electric vehicles, and continuous-duty fluid pumps in industrial factories—permanent magnets must resist catastrophic demagnetization at highly elevated operational temperatures. In this specialized domain, the binary Manganese-Bismuth (MnBi) alloy stands out as an exceptional, highly coveted anomaly in solid-state physics. Unlike almost all other known permanent magnets (including NdFeB), which suffer rapid and severe degradation of their magnetic properties as temperature increases, MnBi features a positive temperature coefficient of coercivity. Incredibly, it is the only known functional permanent magnet material whose coercivity actively *increases* with temperature, nearly doubling its room-temperature resistance to demagnetization as the operating environment approaches 120 °C.³⁸ The theoretical maximum energy product of the functional Low Temperature Phase (LTP) of MnBi is roughly 20 MGOe, driven by a high magnetocrystalline anisotropy energy (1.6×10^6 J/m³).³⁹

4.1 Solving Avalanche Demagnetization via Microstructural Isolation

Despite its extraordinary thermal promise, translating MnBi from a laboratory powder into a commercially viable bulk magnet has been fraught with metallurgical failure for decades. Historically, attempts to consolidate MnBi powders into dense macroscopic magnets resulted in the catastrophic destruction of the material's properties, with up to 70% of the intrinsic coercivity lost during the high-pressure densification process.⁴⁰ Advanced micromagnetic modeling revealed that the coercivity mechanism in bulk MnBi is nucleation-controlled.⁴⁰ At the microscopic level, a bulk magnet is composed of millions of tiny, crystalline grains. When these grains are physically compacted together without a barrier,

inter-grain magnetic exchange coupling occurs.⁴⁰ In this coupled state, if a single defective or misaligned grain undergoes magnetic domain reversal, that reversal cascades through the highly conductive grain boundaries like an avalanche, destroying the macroscopic coercivity of the entire surrounding volume.⁴⁰ To preserve coercivity, the grains must be physically dense but magnetically isolated.

In 2024 and culminating in a prestigious 2025 R&D 100 Award, a breakthrough solution was engineered by scientists Jun Cui and Wei Tang at the U.S. Department of Energy Ames National Laboratory, working under the Critical Materials Innovation Hub.³⁸ To achieve a fully dense, highly anisotropic bulk magnet without sacrificing coercivity, the team developed a specialized inorganic, polymer-based coating applied precisely at the nanoscale to highly purified, finely milled MnBi single-crystal powders.³⁸

This delicate, ultra-thin coating functions as a continuous, non-magnetic grain boundary phase (GBP). It perfectly isolates every individual microscopic grain from its neighbors, even when the powder is subjected to extreme mechanical compaction during the final densification phase.³⁸ Because the grains are physically shielded and magnetically decoupled, the propagation of domain reversal is physically halted at the boundary of any defective grain. Furthermore, during the fabrication and compaction process, a powerful external magnetic field is applied to strictly align the orientation of the isolated particles. This alignment creates a highly anisotropic architecture, ensuring the magnet possesses a preferred direction of magnetization to maximize overall magnetic flux in the target application.³⁸

The resulting rare-earth-free MnBi bulk magnet operates exceptionally well at temperatures up to 200 °C.⁴¹ While some absolute magnetic force is sacrificed to accommodate the non-magnetic polymer coating, this tradeoff is precisely optimized for industrial motors that require high efficiency and extreme temperature stability, but do not strictly necessitate the maximum brute strength of NdFeB.⁴¹ By substituting critically endangered rare earths with bismuth—a cheap, globally abundant element frequently generated as an industrial byproduct of conventional lead and copper smelting—this development establishes a highly economical, resilient supply chain for heavy industry.⁴¹

4.2 Harnessing High-Z Elements: The Discovery of MnBi\$₂ and Co-Bi Phases

The unique thermal and coercive properties of MnBi rely heavily on the quantum mechanical properties of the heavy element Bismuth (Bi), which possesses a very high atomic number (Z=83). In quantum physical terms, high-Z elements generate massive spin-orbit coupling (SOC), an intense relativistic interaction between a bound electron's spin state and its orbital motion around the atomic nucleus.⁴³ When a high-Z element is hybridized with a 3d transition metal that supplies a robust magnetic spin moment (such as Mn or Co), the massive heavy p-orbitals of Bismuth impart a powerful, highly rigid directional preference to the overall magnetic moment.⁴³ This intense directional locking drastically amplifies the magnetocrystalline anisotropy of the entire lattice.⁴³

Recognizing this fundamental property, materials scientists began aggressively exploring the

broader transition metal-Bismuth binary composition space to discover entirely new classes of magnets. This endeavor yielded spectacular results in 2025, driven by a collaborative research consortium including Danna E. Freedman (MIT), Alison B. Altman (Texas A&M), and James M. Rondinelli (Northwestern University).

In research published in the *Journal of the American Chemical Society* (JACS), the team successfully identified and confirmed a novel, previously unknown magnetic phase within the manganese-bismuth system: $\text{MnBi}_{1/2}$.⁴³ Because the newly discovered $\text{MnBi}_{1/2}$ compound decomposes violently upon decompression to ambient atmospheric pressure, its magnetic properties could not be elucidated *ex situ*.⁴³ To overcome this, the researchers utilized a highly specialized diamond anvil cell to achieve ultra-high-pressure synthesis, and then probed the material's magnetism *in situ* using synchrotron X-ray magnetic circular dichroism (XMCD).⁴³ The resulting XMCD spectra definitively proved that $\text{MnBi}_{1/2}$ exhibits strong ferromagnetic hysteresis at both cryogenic (10 K) and ambient room temperatures.⁴³ Advanced computational orbital modeling confirmed that both the Bi p and d orbitals directly hybridize with the Mn lattice to lock the magnetic domains into a highly anisotropic state.⁴³

Following this triumph, the exact same consortium shifted their focus to the Cobalt-Bismuth (Co-Bi) system.⁴⁶ Utilizing advanced computational techniques, specifically *ab initio* random structure searching (AIRSS), the team scoured the thermodynamic landscape for novel structural motifs that might form under extreme pressures.⁴⁶ Transitioning directly from predictive theory to experimental validation, the team successfully synthesized previously

unknown β -CoBi and β - $\text{CoBi}_{1/2}$ phases via high-pressure methods, representing an extraordinary 60% success rate for their computational search algorithms.⁴⁶

Theoretical modeling of these newly stabilized high-pressure phases predicts robust

ferromagnetism.⁴⁶ Crucially, the β -CoBi phase is predicted to possess a magnetocrystalline anisotropy energy vastly superior to traditional permanent magnets, exceeding the intrinsic anisotropy of both expensive CoPt alloys and standard NdFeB magnets.⁴⁶ This establishes the high-Z binary intermetallic framework as one of the most promising frontiers for the next generation of rare-earth-free super-magnets, validating the premise that massive spin-orbit coupling can be synthetically engineered without utilizing f-block elements.

Innovation: High-Temperature Bismuth-Based Magnetics

Material/Technology: MnBi Bulk Magnet Consolidation & Discovery of $\text{MnBi}_{1/2}$ / β -CoBi

Scientific Summary: 1) Engineered scalable MnBi bulk magnets operating up to 200°C by utilizing nanoscale polymer coatings to magnetically isolate grains, preventing avalanche demagnetization during compaction. 2) Utilized High-Z Bismuth

to induce extreme spin-orbit coupling, discovering and synthesizing novel MnBi β and β -CoBi phases under high pressure, with β -CoBi predicted to exceed NdFeB anisotropy.

Key Metrics: MnBi operational limits extended to >200°C; 60% computational-to-experimental success rate for novel high-pressure Co-Bi phases.

Principal Investigators: Dr. Jun Cui (Ames Lab), Prof. Danna Freedman (MIT), Prof. Alison Altman (Texas A&M), Prof. James Rondinelli (Northwestern Univ.)

Publications /

DOIs:(<https://www.ameslab.gov/news/scientists-develop-rare-earth-free-magnet-for-use-in-industrial-motors>)⁴¹, [10.1021/jacs.5c06874](https://doi.org/10.1021/jacs.5c06874)⁴³, [10.1021/jacs.5c12987](https://doi.org/10.1021/jacs.5c12987)⁴⁶

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5. Phase Stabilization and Defect Engineering in Manganese-Aluminum (MnAl)

The binary Manganese-Aluminum (MnAl) system offers another highly compelling rare-earth-free alternative. MnAl is immensely attractive due to its reliance on extremely cheap, globally abundant raw materials, its low specific density (making it ideal for aerospace and automotive weight reduction), and a highly favorable theoretical energy product calculated at approximately 12.64 MGOe.⁴⁹ The magnetic properties of the MnAl system arise exclusively from its metastable, ferromagnetic τ -phase, which, much like tetrataenite, possesses a highly anisotropic $L1_0$ tetragonal crystal structure.⁴⁹

The central metallurgical challenge governing the MnAl system is phase control. The critical, magnetic τ -phase is fundamentally metastable. It forms via a complex, massive displacive-diffusional phase transformation from a non-magnetic, high-temperature hexagonal ϵ -phase.⁴⁹ The window for capturing this phase is incredibly narrow and strictly dependent on precise cooling rates (typically between 10 to 60 K/min).⁵¹ If the alloy is cooled too slowly, the system thermodynamically degrades into highly stable, non-magnetic equilibrium phases (β -Mn and γ_2 -Al), destroying the magnetic potential. Conversely, if the system is quenched too rapidly, the non-magnetic ϵ -phase is entirely retained at room temperature.⁵¹ Recent advancements published extensively in the *Journal of Magnetism and Magnetic*

Materials in late 2025 by researchers Sourav Mandal and Professor Perumal Alagarsamy at the Indian Institute of Technology (IIT) Guwahati demonstrate a highly refined, reproducible approach to capturing this phase and tuning its hard magnetic properties.⁴⁹

By optimizing advanced metallurgical techniques, specifically strip-casting and melt-spinning paired with low-energy, surfactant-assisted milling, the researchers have precisely defined the optimal processing windows for near-equiatomic Mn₅₄Al₄₆ alloys.⁴⁹ The 2025 study highlighted the critical role of intense nanostructuring and deliberate microstructural defect engineering in maximizing the alloy's coercivity.⁴⁹

Employing rapid thermal annealing coupled with the extreme pressures of Spark Plasma

Sintering (SPS), the team successfully orchestrated the massive $\epsilon \rightarrow$ phase transformation while actively suppressing detrimental grain growth.⁴⁹ The SPS process, combined with severe hot deformation, forces the material to undergo high strain, generating a dense network of microstructural defects within the crystalline lattice. Mandal and Alagarsamy definitively identified that the primary mechanism for coercivity enhancement in these bulk structures is domain-wall pinning.⁴⁹ The actively moving magnetic domain walls become physically snagged and locked in place by the deliberately induced structural defects, specifically twin boundaries, shear antiphase boundaries, and high-density dislocations.⁴⁹

The optimized bulk MnAl samples subjected to this rigorous thermo-mechanical processing yielded a robust coercivity of 0.18 T (1.8 kOe) and an impressive saturation magnetization of 114 Am²/kg (114 emu/g).⁴⁹ This saturation magnetization represents a remarkable efficiency,

capturing roughly 79% of the theoretical maximum saturation value (144 emu/g) of the pure τ -phase.⁴⁹ These highly reproducible performance metrics indicate that advanced thermo-mechanical processing can successfully stabilize the elusive MnAl system, positioning it as a highly viable "gap magnet" capable of bridging the performance void between low-cost, low-performance commercial ferrites and highly expensive, critical-mineral-dependent NdFeB architectures.⁴⁹

Innovation: Manganese-Aluminum (MnAl) Microstructural Engineering

Material/Technology: Optimization of the τ -MnAl Phase via Spark Plasma Sintering & Defect Pinning

Scientific Summary: Defined the exact processing windows for the massive $\epsilon \rightarrow$ transformation in Mn₅₄Al₄₆ alloys using melt-spinning, surfactant-assisted milling, and Spark Plasma Sintering (SPS). Achieved high coercivity by deliberately engineering dense networks of microstructural defects (twin boundaries, shear antiphase boundaries) to act as domain-wall pinning sites.

Key Metrics: Achieved 0.18 T (1.8 kOe) coercivity and 114 Am ² /kg (114 emu/g) saturation magnetization (79% of theoretical max).
Principal Investigators: Prof. Perumal Alagarsamy, Sourav Mandal (Indian Institute of Technology Guwahati)
Publications / DOIs: 10.1016/j.jmmm.2025.173410 ⁴⁹
Contact Information: Prof. Perumal Alagarsamy

6. Breaking Symmetry: The Paradox of High-Entropy Magnetic Alloys (HEMAs)

In the realm of advanced structural materials, High-Entropy Alloys (HEAs)—materials composed of five or more principal metallic elements mixed in near-equiatomic proportions—have revolutionized metallurgy. HEAs leverage extraordinarily high configurational entropy to stabilize single-phase solid solutions that exhibit exceptional mechanical properties, such as extreme fracture toughness, high yield strength, and malleability, vastly outperforming traditional binary alloys.⁵⁸

However, transferring this highly successful paradigm into the realm of permanent magnetism presents a profound fundamental paradox. The stabilization mechanism of traditional HEAs inherently favors highly symmetric, chemically disordered crystal structures, overwhelmingly tending to form face-centered cubic (fcc) or body-centered cubic (bcc) lattices.⁵⁹ In magnetic physics, high crystallographic symmetry is highly detrimental to anisotropy. The spherical symmetry of cubic lattices yields exceedingly weak spin-orbit coupling, drastically minimizing magnetocrystalline anisotropy. Consequently, despite their complex elemental makeup, multi-principal element magnetic HEAs are almost universally magnetically "soft," easily flipping their domains and exhibiting meager coercivities on the order of only 10 Oersteds (Oe).⁶⁰

A paradigm-shifting solution to this paradox was detailed in a 2025 study published in *Advanced Materials* by a research team at Georgetown University led by Professor Kai Liu.⁵⁹ The researchers recognized that to achieve the high magnetic anisotropy necessary for a permanent magnet within the vast HEA composition space, they had to aggressively and physically break the cubic symmetry favored by maximum entropy.⁵⁹

6.1 Uniaxial C16 High-Entropy Borides and $L1_0$ HEAs

To achieve this symmetry breaking, Liu's team engineered an entirely novel class of quinary (five-element) high-entropy transition metal borides.⁵⁸ The introduction of interstitial boron atoms into the multi-element matrix dramatically altered the overall thermodynamic landscape. The boron disrupted the preferred cubic packing, forcing the highly complex transition metal matrix to adopt an ordered, highly uniaxial C16 crystal structure.⁶⁰

Once the uniaxial structure was forced, the team utilized a high-throughput combinatorial sputtering approach to systematically explore and tune the magnetic 3d transition metals within the matrix. Specifically, by precisely adjusting the mixing ratios of Iron (Fe) and Cobalt (Co) residing on the transition metal sublattice within the C16 framework, the researchers were able to manipulate the quantum mechanical orientation of the magnetic moments.⁶⁰ This meticulous atomic tuning successfully switched the magnetic easy-axis of the material from a detrimental easy-plane orientation to a strict easy-axis orientation, fundamentally unlocking massively strong magnetocrystalline anisotropy.⁶⁰

Experimental observations of these high-entropy boride films demonstrated a spectacular, more than two-fold increase in raw coercivity when compared against standard binary and ternary transition metal boride baselines.⁶⁰ Highly advanced Density Functional Theory (DFT) calculations corroborating the experimental data indicate that the optimized electronic structure of these specific high-entropy borides can yield an internal anisotropy energy

approaching 10^7 erg/cm³.⁶⁰

Furthermore, in parallel research published in *Advanced Science* (2024), Professor Liu's team demonstrated a second pathway to high anisotropy in HEAs. The researchers deposited thin films of the complex alloy FeCoNiMnCu onto oxidized silicon substrates. Initially, these multi-element films exhibited standard soft magnetic properties with a coercivity of only ~10 Oe.⁶¹ However, the researchers discovered that by incorporating a heavy element (specifically 50 atomic percent Platinum) into the film, and subjecting the material to a highly aggressive Rapid Thermal Annealing (RTA) process, the extreme thermal gradients forced the

multi-element alloy to order into a single, highly aligned $L1_0$ intermetallic phase.⁶¹ This aggressive thermal ordering disrupted the cubic symmetry and achieved an almost 40-fold increase in coercivity over the as-deposited soft films.⁶¹

This dual-pronged approach—utilizing interstitial boron insertion to force uniaxial C16

structures, and utilizing RTA to drive $L1_0$ ordering in heavy-element matrices—establishes the multi-principal high-entropy material space as a remarkably fertile domain. It proves that with extreme elemental engineering, the high-entropy paradigm can indeed be manipulated to discover entirely new classes of highly anisotropic, earth-abundant permanent magnets.

Innovation: High-Entropy Magnetic Alloys (HEMAs)

Material/Technology: C16 Phase High-Entropy Borides & $L1_0$ Ordered HEAs via Rapid Thermal Annealing

Scientific Summary: Solved the cubic-symmetry paradox of HEAs, which typically renders them magnetically soft. By injecting interstitial boron into a quinary transition metal matrix, the team forced the structure into a uniaxial C16 phase. Precise Fe/Co

tuning flipped the magnetic moment to an easy-axis, generating immense anisotropy (10^7 erg/cm ³). Additionally, Rapid Thermal Annealing (RTA) of heavy-element HEAs forced $L1_0$ ordering, yielding a 40-fold coercivity increase.
Key Metrics: Coercivity multiplier of 40x via RTA; theoretical anisotropy approaching 10^7 erg/cm ³ in C16 borides.
Principal Investigators: Prof. Kai Liu, Dr. Willie Beeson (Georgetown University)
Publications / DOIs: 10.1002/adma.202516135 ⁵⁸ , 10.1002/adv.202308574 ⁶¹
Contact Information: Prof. Kai Liu

7. Conclusions and Strategic Outlook

The hyper-accelerated evolution of rare-earth-free permanent magnet research between the years 2024 and 2026 marks a definitive inflection point in global materials science. The collective breakthroughs documented in this exhaustive report represent a fundamental transition of critical-mineral-free (CMF) magnetism from highly theoretical metallurgical models into validated, experimentally proven, and industrially scalable architectures. Several profound insights and actionable conclusions emerge from the synthesis of this data:

1. The Universal Mandate of Symmetry Breaking: The fundamental physics governing the entire endeavor to replace rare-earth elements revolves entirely around the artificial induction of magnetocrystalline anisotropy. Without the inherently anisotropic 4f electron orbitals of elements like Neodymium and Dysprosium, researchers must manipulate the 3d electrons of abundant transition metals. The scientific consensus, validated across multiple distinct material classes, definitively points to structural symmetry breaking as the ultimate requirement.

Whether this asymmetry is achieved through the forced $L1_0$ crystallographic ordering of tetrataenite and MnAl¹⁴, the massive interstitial lattice expansion observed in iron nitrides and high-entropy C16 borides³³, or the extreme spin-orbit coupling artificially introduced by heavy, high-Z elements like Bismuth in Co-Bi systems⁴³, breaking cubic symmetry is the absolute prerequisite for high coercivity.

2. The Ascendancy of Advanced Non-Equilibrium Processing over Pure Chemistry: The early 2020s were largely characterized by highly optimistic attempts to find a simple "magic bullet" chemical dopant to solve severe kinetic diffusion barriers, an era that culminated in the highly publicized, but ultimately retracted, 2022 phosphorus-tetrataenite studies.⁹ The verified, peer-reviewed successes of 2025 and 2026 demonstrate a profound paradigm shift: the future of CMF magnets lies in highly controlled, extreme non-equilibrium thermo-mechanical processing. The utilization of simultaneous stress-field annealing¹⁴, extreme precision in

water-vapor suppression via CaH_2 to control exact nanoparticle growth limits³⁵, nanoscale polymer isolation to prevent magnetic domain avalanche during densification⁴¹, and the aggressive application of Spark Plasma Sintering to induce domain-pinning microstructural defects⁴⁹ all signify that advanced processing technology has eclipsed simple metallurgical alloying as the primary driver of innovation.

3. The Validation of Cross-Disciplinary Integration: The unprecedented intersection of astrophysics, planetary geology, and applied terrestrial metallurgy is yielding profound technological dividends. The confirmation of impact-induced monotectoid tetrataenite formation in the highly pristine Chang'E-6 lunar regolith not only resolves a decades-old planetary magnetic anomaly regarding the lunar farside but provides terrestrial engineers with empirical, macro-scale evidence that high-energy kinetic shock environments can instantly bypass the million-year cooling barriers previously thought insurmountable.²⁵ This cosmomimetic data provides the theoretical underpinning for massive future investments in terrestrial shock-compaction manufacturing pipelines.

4. Commercial Readiness and Stratified Application Integration: As the geopolitical and environmental impetus to decouple from Chinese-dominated critical rare-earth supply chains intensifies³, the industrial integration of these new technologies appears imminent, evolving into a stratified deployment model. The submicron interstitial iron nitride ($\alpha''\text{-Fe}_{16}\text{N}_2$) and the highly defect-engineered MnAl systems are rapidly maturing into the defined "gap magnet" space. They are uniquely poised to dominate the manufacturing of consumer electronics, high-fidelity acoustics, and lower-temperature (<150 °C) continuous-duty appliance motors, driven by their incredible saturation limits and cheap precursor materials.³¹ Concurrently, the breakthrough in polymer-isolated, highly dense MnBi consolidation has definitively secured a viable, rare-earth-free pathway for the extreme high-temperature (>200 °C) environments characteristic of heavy industrial fluid pumps and electric vehicle traction motors.⁴¹

By the close of 2026, the foundational scientific groundwork has unequivocally proven that the intrinsic magnetic limitations of abundant transition metals can indeed be overcome through extreme thermodynamic manipulation and quantum engineering. The discoveries cataloged herein pave the immediate way for a fully sustainable, ecologically sound, and geopolitically resilient global magnetic economy.

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